

Formal Models and Scalable Intelligence in Decentralized Swarm Systems

Information flow, robustness, control architecture, and scaling limits

Tier 3 - Doctoral Research Agenda, revised as a falsifiable long-horizon program

Decentralized Swarm Emergence Research Program

Stage One revision - September 2026

Evidence status: Stage One validates the experimental environment, instrumentation, reward accounting, heuristic baselines, and reproducibility pipeline. It does not yet establish learned emergence, a critical alpha threshold, learned communication, or a phase transition.

Abstract

This doctoral agenda extends the reward-topology and communication-budget studies into a broader theory-and-systems program for decentralized intelligence under scale and uncertainty. The central object is a time-varying interaction graph whose agents act from partial observations under constrained communication, latency, packet loss, energy limits, and failure. Rather than presuming decentralized control eventually surpasses centralized control, the program estimates the regimes in which centralized, decentralized, and hybrid architectures differ in performance, robustness, information efficiency, and scaling cost.

1. Research program

The proposed empirical phase space can be summarized as $E(\alpha, b, N, \xi)$, where α controls reward sharing, b communication capacity, N swarm size, and ξ represents environmental/system uncertainty such as latency, loss, observation noise, and failure. This notation is a research scaffold, not a claim that emergence is reducible to a single scalar function.

2. Core questions

- When does coordination learned at small N transfer or fail as N increases?
- How does communication cost scale with swarm size and graph density?
- Under what perturbations do decentralized or hybrid controllers degrade more gracefully than centralized alternatives?
- Can information-efficiency measures predict performance collapse or resilience?
- Which observed thresholds are robust across task families rather than artifacts of one environment?

3. Comparative control architectures

Architecture	Information access	Primary risk
Centralized	global or near-global state	bottleneck, latency, single-point failure
Decentralized	local observation + constrained messages	partial information, coordination failure
Hybrid/hierarchical	local autonomy + limited aggregation	coordination overhead and hierarchy failure

4. Scale and perturbation matrix

- Swarm sizes extending from validated small- N settings toward 50-100+ agents only as computational and methodological controls permit.
- Latency and packet-loss sweeps.
- Observation noise and distribution-shifted environments.

- Random and structured agent removal.
- Bandwidth and energy budgets.
- Static versus dynamic interaction graphs.

5. Evidence standards

Large-scale claims require matched evaluation budgets, multiple independent training seeds, held-out environment distributions, uncertainty intervals, effect sizes, and sensitivity analysis. Thresholds must be estimated with uncertainty rather than selected visually. Robustness must be measured as degradation and recovery, not inferred from successful anecdotes.

6. Theoretical direction

The program connects MARL with distributed computation, graph dynamics, information theory, and networked control. Formal work may study bounds on information requirements, stability under graph changes, and performance-information tradeoffs. The empirical laboratory remains a testbed; simulation results alone do not establish universal laws of collective intelligence.

7. Long-horizon contribution

A successful doctoral program would deliver a validated hierarchy of empirical models linking reward alignment, information flow, scale, and robustness, together with reproducible software and datasets. The strongest possible outcome is not a predetermined 'critical scale' but a defensible map of where different coordination architectures succeed, fail, and exchange performance for information and resilience.

Figures

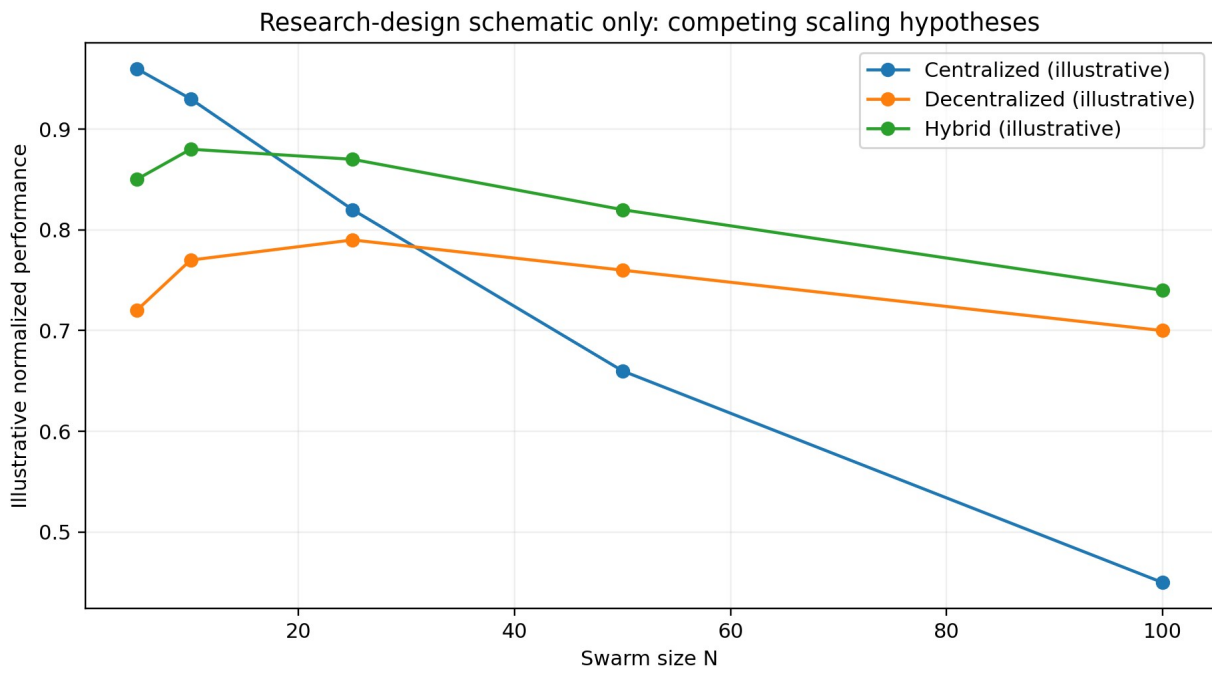


Figure 1. Research-design schematic only. Values are illustrative competing scaling hypotheses, not empirical findings.